

## Design of a Question-Answering Automation System Using an Ontology-Based REG-RAG Architecture

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### Abstract

This study aims to design a question-answering automation system for university academic administration using an Ontology-based REG-RAG architecture. The authors' intention was to reduce unsupported responses generated by large language models (LLMs) and to improve evidence-based decision-making in academic administration queries. To achieve this purpose, four workflows were designed and compared by incrementally combining LLM, RAG, Ontology, and REG components. The ontology structure was used to convert user queries into intent, entity, attribute, condition, time constraint, and required data sources. REG was designed as a rule-based evaluation and guidance layer that selects appropriate reference data sources, rewrites retrieval queries, and constrains response generation. RAG was then used to retrieve evidence from academic schedules, scholarship rules, university regulations, and notices and to provide the evidence to the LLM. The experiment was conducted using K University's academic administration data and evaluated using DCR, DRR, SP, E2E-RT, and SA.

The proposed architecture can reduce arbitrary response generation by the LLM and guide responses based on explicit academic administration data. Therefore, this study suggests that the Ontology-based REG-RAG architecture can contribute to improving decision accuracy and response reliability in academic administration question answering.

**Keywords:** LLM, RAG, Ontology, REG, Academic Administration Automation, Question-Answering System

## 1. Introduction

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Manuscript Received: June. 30, 2026 / Revised: July. 10, 2026 / Accepted: July. 15, 2026

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Recently, university academic administration services have been operated based on various information resources, such as academic calendars, university regulations, scholarship rules, and notices, while students continuously raise repetitive and condition-dependent queries regarding leave of absence, return to school, course registration, scholarship applications, graduation requirements, and related matters. Such queries often require consideration of multiple conditions, including enrollment status, grade criteria, completed credits, application periods, and relevant regulations, rather than merely retrieving information. Therefore, an academic administration question-answering system requires not only the ability to understand natural-language queries but also an architecture capable of selecting appropriate data sources and making evidence-based decisions.

Large language models (LLMs) can be applied to the automation of academic administration question answering based on their natural-language understanding and response-generation capabilities. However, because the LLM-only approach does not directly refer to external academic administration data or the latest notices, it may generate responses that differ from actual regulations or provide answers with unclear evidence. In academic administration queries, dates, application periods, eligibility conditions, and regulatory standards are critical; therefore, such errors are not merely issues of wording but may lead to incorrect administrative decisions. Accordingly, this study regards a structural approach as necessary for reducing hallucination, in which an LLM generates responses without sufficient evidence.

The RAG-based question-answering approach retrieves relevant information from external documents or databases and then provides it to the LLM to generate evidence-based responses. However, general RAG approaches often use the user's natural-language query directly for retrieval; therefore, if query intent and retrieval targets are not clearly distinguished, inappropriate documents may be retrieved. In other words, RAG can mitigate hallucination more effectively than the LLM-only approach, but inaccurate responses may still be generated when incorrect evidence is retrieved or condition-based judgment is omitted.

To address these problems, this study proposes a question-answering architecture that combines ontology-based query structuring with REG (Rule-based Evaluation and Guidance). The ontology structures user queries into intent, entities, attributes, conditions, temporal constraints, and related elements, thereby clarifying the retrieval targets and the meaning of the query. Based on these structuring results, REG selects data sources appropriate for the query type, rewrites retrieval queries, and guides the LLM's response generation so that the retrieved evidence is reflected in the final answer. In this study, REG is defined not as a complete rule-inference engine that independently determines all academic administration conditions, but rather as a rule-based layer that controls retrieval paths and response-generation conditions between the ontology and RAG.

The purpose of this study is to compare four workflows in which RAG, Ontology, and REG are incrementally integrated into an LLM-based question-answering architecture, and to examine the effects of ontology-based query structuring, RAG-based evidence retrieval, and REG-based retrieval control and response guidance on the decision accuracy, response stability, and answer specificity of academic administration question answering. To this end, an identical query set was constructed using K University's university regulations, academic calendar, scholarship rules, and notice data, and the response performance of each workflow was evaluated based on DCR, DRR, SP, E2E-RT, and SA.

## **2. Related Work**

### **2.1 Ontology-based Query Processing**

An ontology is a knowledge-representation method that structurally represents the concepts, attributes, and relationships used in a specific domain. In question-answering systems, ontologies help interpret natural-language queries by separating query intent from target concepts, rather than processing them as simple keyword-based sentences. Previous studies have shown that query understanding can be improved by identifying entities and attributes in user queries and linking them to structured knowledge elements [1].

In ontology-based query processing, user queries are decomposed into elements such as entities, attributes, and relationships and then connected to a domain knowledge structure [1]. This is related to the approach used in this study, in which academic administration queries are decomposed into `primary_intent`, `domain`, `entity`, `attribute`, `condition`, and `time_constraint`.

The academic administration domain includes various subdomains, such as academic calendars, university regulations, scholarship rules, and notices. Therefore, even similar expressions may require different data sources depending on the query intent. A topic-map-based retrieval study on university academic administration records suggested that retrieval efficiency can be improved by structuring academic administration information into business functions and detailed topics [2]. In addition, Park and Kim designed an ontology-based college information retrieval system by structuring college, graduate school, and member information, showing that university-related information can be organized into concepts, relationships, and attributes [3]. Based on these studies, this study uses an ontology as a semantic structure for interpreting academic administration queries and selecting appropriate data sources.

## 2.2 Retrieval-Augmented Generation-based QA System

The original RAG framework combines the parametric knowledge of a pre-trained generative model with non-parametric knowledge retrieved from external sources, allowing knowledge-intensive NLP tasks to use external evidence during generation [4]. Recent RAG research has expanded into various architectures, and a survey classified RAG into Naive RAG, Advanced RAG, and Modular RAG, emphasizing retrieval, generation, and augmentation as core components [5].

However, general RAG approaches often use the user's natural-language query directly for retrieval, which may lead to inaccurate retrieval when query intent and data sources are not clearly distinguished. To address this issue, ontology-grounded RAG structures factual relationships using domain ontology to improve the reliability of retrieval evidence [6]. Metadata-based RAG has also been proposed to improve answer accuracy and evidence traceability by incorporating metadata such as document titles, pages, and keywords into embedded data and by requiring source information during prompt construction [7].

In the academic administration domain, RAG-based question-answering systems require document parsing, preprocessing, and retrieval-structure optimization because academic data exist in various formats, including PDF files, HWP files, notices, and regulation documents [8]. Therefore, this study compares a simple RAG approach with an approach that combines ontology-based query structuring, REG-based retrieval control, and response guidance.

## 2.3 Rule-based Evaluation and Guidance (REG)

In domains where explicit evidence exists, such as regulations, systems, and standards, generative models may produce responses that are natural but inconsistent with actual rules. Academic administration queries

often require simultaneous consideration of regulations, schedules, notices, and user conditions, such as scholarship eligibility, course add/drop periods, and leave-of-absence limits. Therefore, retrieving related documents alone is not sufficient; a control procedure is needed to select appropriate data sources and ensure that retrieved evidence is reflected in the final response.

Prior research pointed out that when generative AI directly intervenes in decision-making, the consistency and reproducibility of results may be weakened. It therefore proposed separating judgment from generation using an ontology and a rule-based layer [9]. In addition, ontology- and knowledge-graph-based QA studies have shown that linking natural-language queries to structured knowledge elements can improve query processing. Bio-SODA, for example, links natural-language queries to knowledge-graph elements and generates candidate SPARQL queries without separate training data [10].

Recent RAG variants have also focused on controlling retrieval strategies. Adaptive-RAG dynamically selects no retrieval, single-step retrieval, or multi-step retrieval according to query complexity [11]. Similarly, CRAG introduces a retrieval evaluator that assesses the quality of retrieved documents and triggers corrective actions when retrieval results are inaccurate or ambiguous [12]. These studies support the need for an intermediate control layer that evaluates retrieval paths and guides generation based on evidence quality.

Based on these studies, this study defines REG as Rule-based Evaluation and Guidance. REG is not a complete rule-inference engine that independently determines all administrative conditions. Rather, it is a rule-based control layer that selects retrieval targets, rewrites retrieval queries, and guides LLM response generation based on ontology mapping results. In this study, Evaluation refers to evaluating query type, reference data sources, retrieval paths, and response constraints, while Guidance refers to guiding response generation so that retrieved evidence and query-structuring results are reflected in the final response.

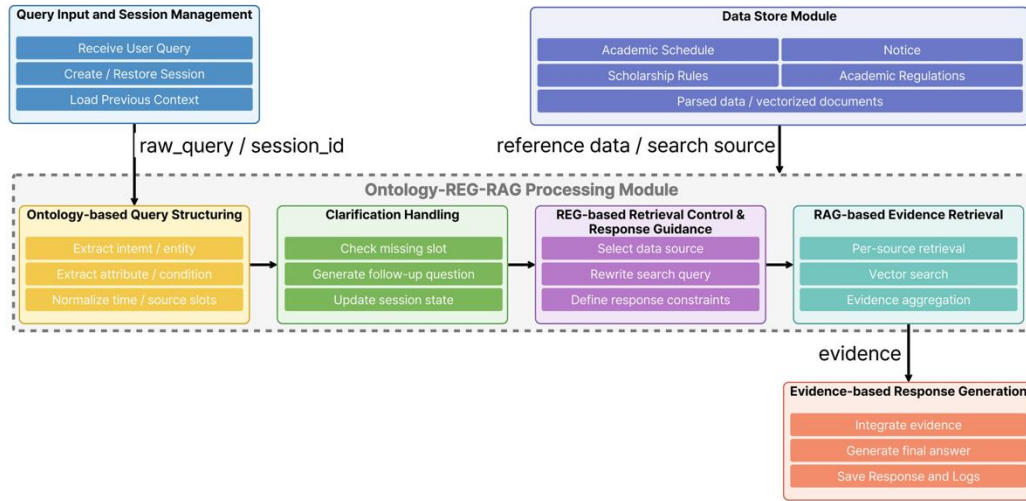
Accordingly, user queries are structured through the ontology into primary\_intent, entity, attribute, condition, time\_constraint, needs\_sources, and related elements. REG then selects appropriate data sources from academic calendars, notices, scholarship rules, and university regulations. Therefore, REG in this study inherits the concerns of rule-based judgment control discussed in prior research [9], but redefines them for the academic administration question-answering environment as a retrieval-control and response-guidance layer between ontology-based query structuring and RAG retrieval.

### **3. System Design and Implementation**

This section describes the design and implementation architecture of the Ontology-REG-RAG-based system for automating university academic administration question answering. The proposed system consists of a Data Store Module, which organizes academic administration data into a storable and searchable form; Query Input and Session Management, which manages user queries and session information; an Ontology-REG-RAG Processing Module, which structures queries and generates retrieval plans and evidence sets; and Evidence-based Response Generation, which generates final responses and logs based on retrieved evidence.

The system proposed in this study does not allow the LLM to respond directly to the user query. Instead, it first normalizes the user query into an ontology structure, requests additional information when necessary, determines the data source and retrieval query through REG-based retrieval control, and then generates the final response based on RAG-based evidence retrieval results. This architecture is intended to reduce problems that may occur in academic administration queries, such as data-source confusion, omitted conditions, and unsupported natural-language generation, and to guide evidence-based response generation appropriate to the

query type.



**Figure 1. Modular Architecture of the Ontology-REG-RAG based Academic Administration Question-Answering System**

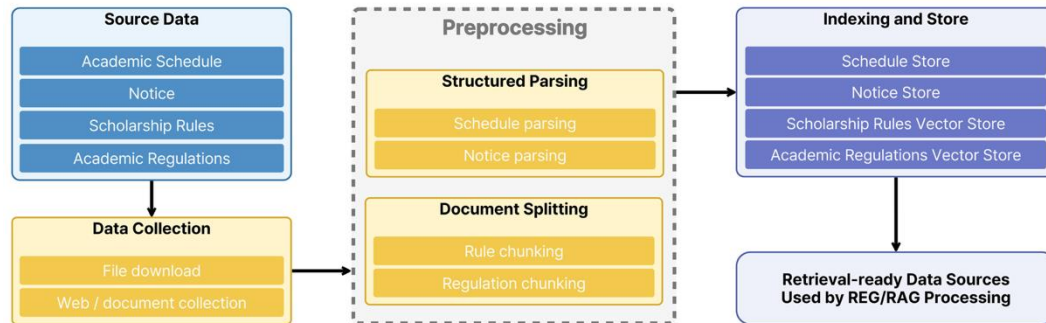
Figure 1 presents the overall modular architecture of the proposed system. User queries are collected by Query Input and Session Management and managed together with session information. Subsequently, the Ontology-REG-RAG Processing Module sequentially performs ontology-based query structuring, clarification handling, REG-based retrieval control and response guidance, and RAG-based evidence retrieval. The Data Store Module manages academic calendar, notice, scholarship rule, and university regulation data; in the REG stage, it is used for selecting reference data sources, and in the RAG stage, it serves as the target for evidence retrieval. Finally, Evidence-based Response Generation integrates the retrieved evidence and structured query information to generate the final response and stores response and evaluation logs.

### 3.1 Data Store Module

The Data Store Module collects, preprocesses, and indexes reference data required for academic administration question answering and organizes it into a searchable form. This study used K University's academic calendar, notices, scholarship rules, and university regulations. Depending on their characteristics, these data are divided into structured data and document-based data. Academic calendars and notices include explicit fields such as title, date, start date, end date, and URL, and were therefore processed as structured data. Scholarship rules and university regulations include unstructured document elements such as articles, conditions, and explanatory text, and were therefore used through document chunking and embedding-based similarity retrieval.

Academic calendar data are used for queries requiring date-based judgment, such as leave of absence, return to school, course registration, course add/drop, and examination periods. Notice data are used for queries requiring confirmation of recent information, such as scholarship application announcements, recruitment notices, competitions, and event announcements. Scholarship rule data are used for queries requiring condition verification, including scholarship types, eligible applicants, grade criteria, completed credits, income brackets,

and payment standards, while university regulation data are used for queries requiring confirmation of institutional standards, such as leave of absence, return to school, withdrawal, expulsion, and graduation requirements.



**Figure 2. Data Collection and Indexing Structure of the Data Store Module**

Figure 2 illustrates the data collection and indexing structure of the Data Store Module. The source data consist of academic calendars, notices, scholarship rules, and university regulations, and each data type undergoes a different preprocessing procedure. Academic calendars and notices are parsed around structured fields, whereas scholarship rules and university regulations undergo document chunking and are indexed in a form suitable for vector retrieval. The resulting data store is subsequently used in the REG stage for reference data-source selection and in the RAG stage as the target for evidence retrieval.

The purpose of the Data Store Module is not merely to store documents, but to distinguish academic administration data into semantic units so that an appropriate data source can be selected according to the query type. Accordingly, in this study, the academic calendar, notices, scholarship rules, and university regulations are managed as separate reference units, and each data source is configured so that it can be independently selected in the REG and RAG stages. This separation clarifies which data a user query should be grounded in and thereby contributes to reducing retrieval errors.

### 3.2 Query Input and Session Management

Query Input and Session Management is the stage that collects user queries and manages the basic metadata and session information required for query processing. The user's natural-language query is stored as the `raw_query` and is subsequently used as a common input in the stages of ontology-based query structuring, clarification handling, REG-based retrieval control, RAG-based evidence retrieval, and final response generation.

For the evaluation experiment, this study assigned `scenario_id`, `session_id`, `repeat_run_id`, `reference_label`, and `required_answer_elements` to each query. The `scenario_id` identifies the same basic query scenario, while the `repeat_run_id` is used to evaluate response stability by repeatedly executing the same query. The `reference_label` is the ground-truth label used to evaluate final decision accuracy, and `required_answer_elements` refers to the mandatory detailed elements needed to evaluate answer specificity.

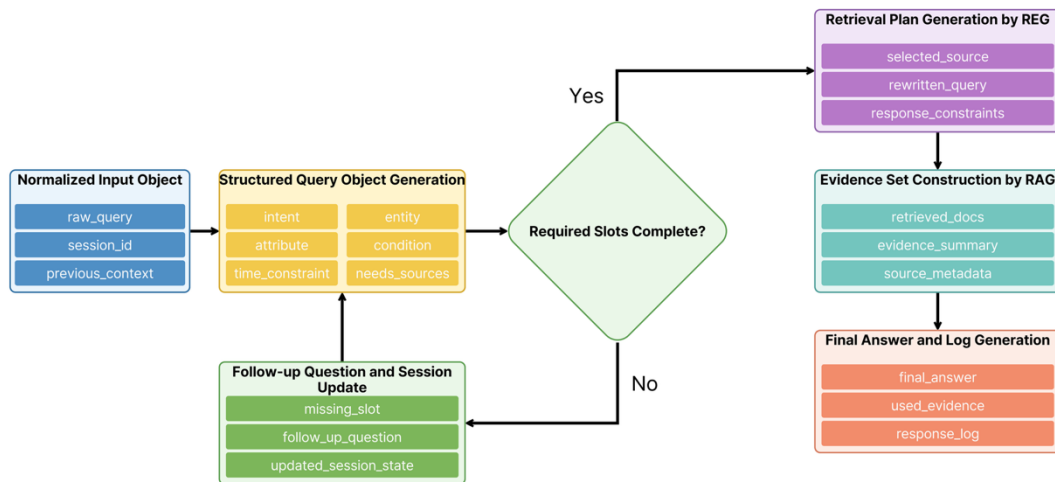
This stage also manages session information so that not only single-turn queries but also queries requiring additional information can be processed. For example, a query such as "Check whether I can apply for an academic performance scholarship" may be classified as a scholarship rule query, but actual response

generation may require additional information such as enrollment status, previous-semester GPA, completed credits, and disciplinary status. In such cases, the system links the previous query with supplementary information based on the `session_id` and uses the accumulated context in the subsequent response-generation stage.

This subsection does not present Query Input and Session Management as a separate implementation screen, but instead focuses on how the input query and session information are used as common inputs in subsequent processing stages. The normalized input object is later transformed into a structured query object in the Ontology-REG-RAG Processing Module, and this process is described in the next subsection as an output-based execution flow.

### 3.3 Ontology-REG-RAG Processing Module

The Ontology-REG-RAG Processing Module is the core processing stage of the proposed system. It structures user queries, determines information sufficiency, and generates retrieval plans and evidence sets. Rather than immediately passing a natural-language query to the LLM for response generation, this module processes the query through an output flow consisting of the Normalized Input Object, Structured Query Object, Retrieval Plan, Evidence Set, and Final Answer and Log. This output-based execution flow is shown in Figure 3.



**Figure 3. Output-based Execution Flow of the Ontology-REG-RAG Processing Module**

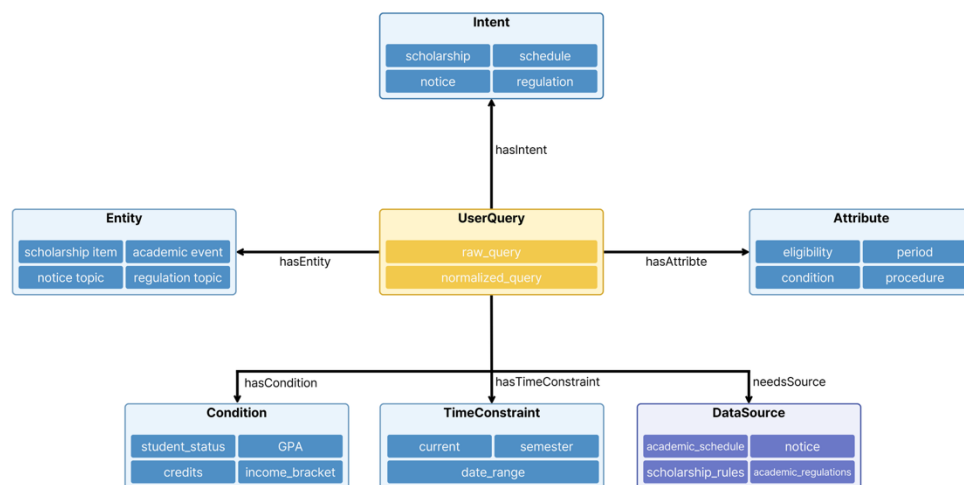
Figure 3 shows the output-based execution flow of the Ontology-REG-RAG Processing Module. The normalized input object consists of `raw_query`, `session_id`, and `previous_context`, and is transformed in the Structured Query Object Generation stage into slots such as `intent`, `entity`, `attribute`, `condition`, `time_constraint`, and `needs_sources`. The system then determines whether the required slots have been filled. If a required slot is missing, the system generates `missing_slot` and `follow_up_question`, reflects them in `updated_session_state`, and passes the supplemented input back to the structuring stage. If all required slots are satisfied, REG generates a retrieval plan consisting of `selected_source`, `rewritten_query`, and `response_constraints`, and RAG constructs an evidence set that includes `retrieved_docs`, `evidence_summary`, and `source_metadata`. Finally, the system generates `final_answer`, `used_evidence`, and `response_log`.

Unlike Figure 1, which presents the overall module structure, Figure 3 shows how a single query is internally transformed into data objects. That is, while Figure 1 explains the relationships among system modules, Figure 3 explains the execution procedure within the Processing Module, in which a query is transformed into a structured object, a retrieval plan, an evidence set, and the final response and log.

### 3.3.1 Ontology-based Query Structuring

Ontology-based Query Structuring is the process of converting a user query into a structured query representation appropriate for the academic administration domain. In this study, user queries were normalized into slots such as `primary_intent`, `secondary_intents`, `entity`, `attribute`, `condition`, `time_constraint`, `needs_sources`, and `user_action`. The `primary_intent` represents the main intent of the query and is classified as one of SCHOLARSHIP, SCHEDULE, NOTICE, RULES, MULTI, and UNKNOWN. The entity denotes the central object of the query, such as an academic performance scholarship, leave of absence, return to school, course add/drop, or competition, while the attribute represents the detailed property requested by the user, such as eligibility, period, condition, criterion, or latest notice. The `time_constraint` expresses temporal conditions such as “current,” “this semester,” “a specific date,” or “date range,” and `needs_sources` denotes the list of data sources that must be referenced to process the query.

In academic administration queries, the ontology is used to transform natural-language queries into intermediate representations that can be used for retrieval and decision making. Even when the same expression is used, academic administration queries may require different data sources depending on whether the query concerns a period, condition, procedure, or regulation. For example, “leave-of-absence application” is linked to academic calendar data when the user asks about the application period, but is linked to university regulation data when the user asks about application eligibility or the number of semesters allowed. Therefore, this study decomposes user queries into Intent, Entity, Attribute, Condition, TimeConstraint, and DataSource to explicitly represent query intent and retrieval targets. These structuring results are used in the REG stage for data-source selection and retrieval-query rewriting, and in the RAG stage as criteria for constraining retrieval targets and evidence scope.



**Figure 4. Academic Administration Query Ontology Structure for Ontology-based Query Structuring**

Figure 4 presents the academic administration query ontology structure used in Ontology-based Query Structuring. A user query is connected to Intent, Entity, Attribute, Condition, TimeConstraint, and DataSource through UserQuery. Intent represents the topic of the query and is classified into categories such as scholarships, academic schedules, notices, and university regulations. Entity refers to the scholarship item, academic schedule item, notice topic, or regulation topic that is the target of the query, while Attribute represents the property the user wants to check, such as eligibility, period, condition, or procedure. Condition includes user conditions required for judgment, such as enrollment status, GPA, completed credits, and income bracket, while TimeConstraint represents temporal constraints such as the current point in time, semester, or date range. DataSource specifies the data sources to be referenced in the subsequent REG and RAG stages.

The core of the ontology design is to transform user queries from simple natural-language sentences into structured inputs for retrieval and response generation. For example, the query “I am currently enrolled; check whether I can apply for an academic performance scholarship” is structured as `primary_intent=SCHOLARSHIP`, `entity=academic performance scholarship`, `attribute=eligibility`, `condition={student_status: enrolled}`, and `needs_sources={scholarship_rules}`. In contrast, the query “Is the course add/drop period still open?” is structured as `primary_intent=SCHEDULE`, `entity=course add/drop`, `attribute=period check`, `time_constraint=current`, and `needs_sources={academic_schedule}`.

The ontology slots and their meanings used in this study are shown in Table 1.

**Table 1. Ontology Slot Design**

| Slot              | Description            | Example Values                                                                   | Purpose of Use                 |
|-------------------|------------------------|----------------------------------------------------------------------------------|--------------------------------|
| raw_query         | Original user query    | Check whether I can apply for an academic performance scholarship                | Preserve the original query    |
| primary_intent    | Main query intent      | SCHOLARSHIP, SCHEDULE, NOTICE, RULES, MULTI, UNKNOWN                             | Select data source             |
| secondary_intents | Secondary query intent | NOTICE, SCHEDULE                                                                 | Process complex queries        |
| entity            | Query target           | Academic performance scholarship, course add/drop, leave of absence, competition | Construct retrieval query      |
| attribute         | Required attribute     | Eligibility, period, condition, criterion                                        | Determine response elements    |
| condition         | User condition         | Enrolled student, GPA, completed credits, income bracket                         | Support condition verification |
| time_constraint   | Temporal constraint    | Current, 2026 Spring semester, this semester                                     | Compare schedules              |
| needs_sources     | Reference data source  | scholarship_rules, schedule, notice, rules                                       | REG retrieval control          |
| user_action       | User-requested action  | Check, retrieve, summarize, judge                                                | Determine response type        |

### 3.3.2 Clarification Handling

Clarification Handling is the process of requesting additional information when the information required for a user query is insufficient. For some academic administration queries, the query intent and target may be clear,

but the condition information required to generate the final response may be lacking. For example, a query about scholarship eligibility may require not only the scholarship name but also detailed conditions such as enrollment status, previous-semester GPA, completed credits, disciplinary status, and income bracket. If the LLM directly determines eligibility or ineligibility while such information is missing, there is a high likelihood that an unsupported response will be generated.

In this study, the satisfaction of required slots is checked based on the results of ontology-based query structuring, and when information is insufficient, `missing_slot` and `follow_up_question` are generated. For example, if the user enters “Check whether I can apply for an academic performance scholarship” but provides neither enrollment status nor grade information, the system can request additional information, such as “To verify the academic performance scholarship, please provide your enrollment status, previous-semester GPA, and completed credits.” This stage reduces judgment errors by preventing incomplete queries from being immediately passed to retrieval or response generation.

Clarification Handling is not applied to all queries; it is performed only when user conditions required for the final response are clearly insufficient. Queries that can be answered using only the original data, such as simple schedule lookup or notice retrieval, are passed to the REG and RAG stages without clarification. Accordingly, Clarification Handling functions as a safeguard that checks the information sufficiency of user queries and obtains the condition information required for judgment-type queries.

### **3.3.3 REG-based Retrieval Control and Response Guidance**

REG-based Retrieval Control and Response Guidance is the stage that controls retrieval targets and retrieval queries based on ontology-based query-structuring results and guides the retrieved evidence to be reflected in the final response. In this study, REG stands for Rule-based Evaluation and Guidance and is defined not as a complete rule-inference engine that independently determines all academic administration conditions, but as a rule-based control module that controls retrieval paths and response-generation conditions between the ontology and RAG.

The Evaluation component of REG determines which data sources should be referenced by evaluating the query type, entity, attribute, temporal constraint, user condition, and `needs_sources`. For example, when the `primary_intent` is `SCHOLARSHIP` and the entity is an academic performance scholarship, REG prioritizes scholarship rule data. If the query also contains expressions such as “application period,” “notice,” or “recruitment,” the retrieval path is expanded to also reference academic calendar or notice data. In contrast, when the `primary_intent` is `RULES`, university regulation data are prioritized, and when it is `SCHEDULE`, academic calendar data are prioritized.

The Guidance component of REG configures response-generation conditions so that the retrieved evidence and query-structuring results are reflected in the final LLM response. For example, in scholarship rule queries, it guides the response to include the scholarship name, eligibility conditions, grade criteria, completed-credit requirements, and additional confirmation items, while in academic schedule queries, it guides the response to include the schedule name, start date, end date, and current availability. In this way, the system controls the LLM so that it does not arbitrarily generate content unrelated to the retrieved evidence, but instead answers around the mandatory response elements for each query type.

Examples of the REG-based retrieval control and response guidance rules applied in this study are shown in Table 2.

**Table 2. Examples of REG Retrieval Control and Response Guidance Rules Based on Ontology Cues**

| Query Type              | Ontology Cue                                                                                     | REG Evaluation                                                                                                              | REG Guidance                                                                                                                                 | Reference Data         |
|-------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Scholarship eligibility | primary_intent=SCHOLARSHIP,<br>entity=academic performance scholarship,<br>attribute=eligibility | Prioritize the scholarship rule source and rewrite retrieval terms for grades, completed credits, and enrollment conditions | Guide the response to include scholarship name, eligibility conditions, grade criteria, completed credits, and additional confirmation items | Scholarship rules      |
| Academic schedule check | primary_intent=SCHEDULE,<br>entity=course add/drop,<br>time_constraint=current                   | Select the academic calendar source and rewrite retrieval terms related to dates and deadlines                              | Guide the response to include schedule name, start date, end date, and current availability                                                  | Academic calendar      |
| Notice retrieval        | primary_intent=NOTICE,<br>entity=competition,<br>attribute=latest/recruitment                    | Select the notice source and rewrite retrieval terms related to latest, recruitment, and application                        | Guide the response to include notice title, posting date, and URL                                                                            | Notices                |
| Regulation query        | primary_intent=RULES, entity=leave of absence,<br>attribute=regulation/condition                 | Select the university regulation source and rewrite retrieval terms related to articles, standards, and requirements        | Guide the response to include the related regulation name, article or evidence content, and applicable conditions                            | University regulations |
| Complex query           | needs_sources is two or more                                                                     | Retrieve from multiple sources and integrate evidence                                                                       | Answer by distinguishing the roles of different evidence items                                                                               | Multiple data sources  |

The REG rules in this study select reference data sources using `primary_intent` as the primary criterion and use `attribute`, `time_constraint`, and `needs_sources` as auxiliary criteria. In the case of complex queries, if `needs_sources` is set to two or more sources, retrieval is performed across multiple sources, and REG Guidance specifies what role the evidence retrieved from each source should play in the final response.

### 3.3.4 RAG-based Evidence Retrieval

RAG-based Evidence Retrieval is the process of retrieving relevant evidence and constructing an evidence set based on the retrieval plan generated by REG. Although general RAG approaches often use the user's natural-language query directly for retrieval, this study uses `rewritten_query` and `selected_source` produced through ontology-based query structuring and REG-based retrieval control. This reduces the likelihood of retrieving documents unrelated to the query type and prioritizes the acquisition of evidence required for the final response.

RAG retrieval is performed differently depending on the data type. Academic calendars and notices can be

retrieved and sorted using structured fields such as title, date, and URL, whereas scholarship rules and university regulations are searched for relevant rules or articles through document chunking and embedding-based similarity retrieval. The retrieval results are organized into `retrieved_docs`, `evidence_summary`, and `source_metadata` and then passed to the final response-generation stage.

Documents retrieved in RAG-based Evidence Retrieval are not simply added to the LLM input, but are used together with the `response_constraints` defined by REG Guidance. For example, in notice queries, the title, posting date, and URL of the retrieved notice must be included in the final response, while in scholarship rule queries, the scholarship name, conditions, grade criteria, completed-credit criteria, and additional confirmation items must be reflected in the response. In this way, the RAG stage secures retrieval evidence and provides a structured evidence set so that an evidence-based response can be generated in the final response-generation stage.

### **3.4 Evidence-based Response Generation and Logging**

Evidence-based Response Generation is the stage that generates the final response and log based on the structured query object, retrieval plan, and evidence set produced by the Processing Module. In the final response-generation stage, the LLM provides a natural-language response to the user, but the response is constrained to be based on retrieved evidence and structured query information. In other words, the LLM is not used as an independent decision maker, but as a generation module that converts retrieved evidence and REG-based response conditions into sentences that the user can understand.

The final response takes different forms depending on the user's query type. Schedule queries include the schedule name, start date, end date, and availability as of the current point in time, while notice queries include the notice title, posting date, and URL. Scholarship rule queries include the scholarship name, eligibility conditions, grade criteria, completed-credit criteria, and items requiring additional confirmation, while university regulation queries include the related regulation name, article or evidence content, and applicable conditions. When condition-based judgment is incomplete, the system does not assert eligibility or ineligibility, but instead presents the information that requires further confirmation.

Evidence-based Response Generation also stores `response_log` together with the final response for experimental evaluation. The `response_log` includes `predicted_label`, `reference_label`, `response_time`, `used_evidence`, `ontology_result`, `reg_applied`, `retrieved_docs`, and related elements. The `predicted_label` is the core decision label generated by the system and is used to calculate DCR by comparing it with the `reference_label`. Repeated execution results for the same query are used to calculate DRR, and the required elements included in the response are used to calculate SP. The ontology-structuring results are used to calculate SA, and the processing time from input to final response is recorded as E2E-RT.

As described above, Evidence-based Response Generation is not merely a stage for returning responses; it simultaneously performs evidence-based response generation and experimental evaluation logging. This enables the study to quantitatively compare the response quality of the proposed system and to analyze how the components of each Flow affect decision accuracy, decision reproducibility, answer specificity, and processing time.

## **4. Experiments**

#### 4.1 Experimental Environment

In this study, four workflows were executed under the same query set and the same data conditions, and their performance was compared. The experiment was conducted in an n8n-based workflow environment using K University's academic calendar, scholarship rules, university regulations, and notice data. The number of evaluation logs for each workflow was 130.

The main components of the experimental environment are shown in Table 3. Table 3 summarizes the execution environment, comparison targets, data used, evaluation method, evaluation metrics, and number of evaluation logs.

**Table 3. Experimental Environment Configuration**

| Category                  | Description                                                                        |
|---------------------------|------------------------------------------------------------------------------------|
| Execution environment     | n8n-based workflow                                                                 |
| Comparison targets        | Flow A, Flow B, Flow C, Flow D                                                     |
| LLM model                 | Gemma 4-based LLM (gemma4-e4b)                                                     |
| Ontology temperature      | 0.1                                                                                |
| Answer temperature        | 0.2                                                                                |
| Embedding model           | Gemini Embedding-series model (gemini-embedding-2)                                 |
| Data used                 | K University academic calendar, scholarship rules, university regulations, notices |
| Evaluation method         | Response comparison by workflow based on the same query set                        |
| Evaluation metrics        | DCR, DRR, SP, E2E-RT, SA                                                           |
| Number of evaluation logs | 130 per workflow                                                                   |

As shown in Table 3, all workflows were evaluated based on the same data and the same query set. RAG retrieval was performed through document chunking and embedding-based similarity retrieval, and this study focused on comparing response performance according to differences in the components of each Flow rather than on a particular data-store implementation. Therefore, differences in the experimental results can be interpreted as arising from differences in workflow components rather than differences in data conditions.

The experimental query set consisted of 26 basic scenarios, and each scenario was executed five times to generate 130 evaluation logs per workflow. The query types consisted of eight scholarship rule queries, five university regulation queries, five academic schedule queries, four notice queries, and four complex queries. Each test case included a scenario\_id, user query, reference judgment label (reference\_label), required answer elements (required\_answer\_elements), repeated execution number (repeat\_run\_id), and session identifier (session\_id). This design enabled comparison of decision accuracy, repeated-execution stability, and answer specificity by Flow for the same queries.

The composition of the experimental query set by type is shown in Table 4. Table 4 classifies the 26 basic scenarios into scholarship rules, university regulations, academic schedules, notices, and complex queries, and summarizes the number of repetitions and evaluation logs for each type.

**Table 4. Composition of the Experimental Query Set**

| Query Type             | Number of Scenarios | Number of Repetitions | Number of Evaluation Logs | Examples                                                                                                                |
|------------------------|---------------------|-----------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Scholarship rules      | 8                   | 5                     | 40                        | Academic performance scholarship eligibility; income bracket for welfare scholarship                                    |
| University regulations | 5                   | 5                     | 25                        | Leave-of-absence procedure; allowable semesters for general leave; grounds for expulsion                                |
| Academic schedules     | 5                   | 5                     | 25                        | Course add/drop period; course registration period; final examination period                                            |
| Notices                | 4                   | 5                     | 20                        | Competition notice; scholarship application notice; employment program notice                                           |
| Complex queries        | 4                   | 5                     | 20                        | Academic performance scholarship conditions and application period; leave-of-absence regulations and application period |
| Total                  | 26                  | 5                     | 130                       | -                                                                                                                       |

In this study, the number of repeated executions  $R$  for calculating DRR was set to 5.

#### 4.2 Evaluation Metrics

This study used DCR, DRR, SP, E2E-RT, and SA as the final evaluation metrics. DCR is the ratio at which the system's final decision result matches the reference answer, and DRR is the ratio at which the same decision is reproduced when the same input is repeatedly executed. SP is the ratio of mandatory detailed elements that should be included in the reference answer and are actually included in the response, and E2E-RT is the average processing time from input to final response. SA evaluates whether a user query is transformed without omission into the main structural elements required in the ontology-structuring stage, and was applied only to Flow C and Flow D, which include the ontology-structuring stage.

Evaluation in this study was performed at the query level. Each evaluation item was constructed by comparing the final response generated by a specific Flow for the same user query with the reference answer. The reference answer ( $y_i^{ref}$ ) was based on the original texts of the academic calendar, scholarship rules, university regulations, and notices. However, an LLM was used to generate draft reference judgment labels and required answer elements for each query, which were then reviewed by the researcher against the original texts. The reference answer was divided into `reference_label` and `required_answer_elements`: `reference_label` was used to calculate DCR and DRR, and `required_answer_elements` was used to calculate SP. Depending on the query type, the reference answer was classified according to categories such as eligibility, whether the period was still open, whether related information existed, and whether evidence was included. For example, in period-judgment queries, the reference answer was based on whether the current date fell between the start and end dates of the relevant schedule, whereas in scholarship-condition queries, it was based on mandatory conditions required by the scholarship rules, such as enrollment status, grade criteria, and completed credits.

The final evaluation metrics used in this study are summarized in Table 5. Table 5 presents the definition,

evaluation perspective, and application target of each metric.

**Table 5. Final Evaluation Metrics Adopted**

| Abbreviation | Evaluation Metric             | Formula                                                                      | Evaluation Perspective                  | Application Target |
|--------------|-------------------------------|------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| DCR          | Decision Correctness Rate     | $DCR = \frac{1}{N} \sum_{i=1}^N I(y_i^{sys} = y_i^{ref})$                    | Final decision accuracy                 | All Flows          |
| DRR          | Decision Reproducibility Rate | $DRR = \frac{1}{N} \sum_{i=1}^N (y_i^{(1)} = y_i^{(2)} = \dots = y_i^{(R)})$ | Repeated-execution stability            | All Flows          |
| SP           | Specificity                   | $SP = \frac{N_{covered\_elem}}{N_{req\_elem}}$                               | Answer specificity                      | All Flows          |
| E2E-RT       | End-to-End Response Time      | $RT_i = t_i^{out} - t_i^{in}$                                                | Processing efficiency                   | All Flows          |
| SA           | Structural Alignment          | $SA = \frac{N_{correct\_struct}}{N_{ref\_struct}}$                           | Generation of major structural elements | Flow C, Flow D     |

As shown in Table 5, DCR, DRR, SP, and E2E-RT were applied to all Flows, while SA was applied only to Flow C and Flow D, which include the ontology-structuring stage. In this study, SA was used not as a metric for evaluating exact agreement with a ground-truth ontology, but as a structural completeness metric for checking whether the major structural elements required for response generation and retrieval control were generated without omission. In addition, this study did not directly measure hallucination as a separate metric; instead, changes in decision errors and answer specificity were indirectly examined through DCR and SP.

DCR evaluates whether the final decision produced by the system matches the reference answer, and was calculated based on the core decision unit of the query rather than the entire natural-language response. For example, a query such as “Is the course add/drop period still open?” was judged as within period/period ended; a query such as “Are there any recent competition-related notices?” was judged as related information present/absent; and a query such as “Check whether I can apply for an academic performance scholarship” was judged as eligible/ineligible/additional information required. DRR evaluated whether the core decision result was reproduced consistently when the same query was executed repeatedly. SP was calculated as the proportion of mandatory detailed elements in the reference answer that were included in the actual response. The required detailed elements were defined differently depending on the query type: for academic schedule queries, the main elements were schedule name, start date, end date, and current availability; for notice queries, notice title, posting date, and URL; and for scholarship rule queries, scholarship name, eligibility conditions, grade or completed-credit criteria, and items requiring additional confirmation.

## 5. Results

### 5.1 Quantitative Evaluation Results

The quantitative evaluation results for the four workflows are shown in Table 6. Table 6 compares the DCR,

DRR, SP, E2E-RT, and SA results for each Flow.

**Table 6. Quantitative Evaluation Results by Workflow**

| Pipeline                     | DCR↑   | DRR↑   | SP↑    | E2E-RT↓ | SA↑  |
|------------------------------|--------|--------|--------|---------|------|
| Flow A: LLM-Only             | 0.3538 | 0.8846 | 0.1335 | 11.19 s | N/A  |
| Flow B: LLM+RAG              | 0.5846 | 0.9615 | 0.8263 | 15.18 s | N/A  |
| Flow C: LLM+RAG+Ontology     | 0.6385 | 0.8462 | 0.8217 | 18.74 s | 1.00 |
| Flow D: LLM+REG+RAG+Ontology | 0.6923 | 0.8462 | 0.8154 | 15.06 s | 1.00 |

As shown in Table 6, the DCR values were 0.3538 for Flow A, 0.5846 for Flow B, 0.6385 for Flow C, and 0.6923 for Flow D, with Flow D achieving the highest final decision accuracy. This indicates a tendency for the proportion of decisions matching the reference answers to increase as RAG, Ontology, and REG are incrementally integrated.

The quantitative results of this study compare performance differences among workflows on the same query set, and their purpose is to examine performance trends according to the integration of components rather than to conduct statistical significance testing.

## 5.2 Analysis of Evaluation Metrics

SP was low in Flow A at 0.1335, but increased substantially to 0.8263 in Flow B. This indicates that the RAG architecture, which provides externally retrieved documents to the LLM, is effective in including the detailed elements required in answers. Flow C and Flow D also showed similar levels, with values of 0.8217 and 0.8154, respectively, indicating that evidence retrieval and query structuring contributed to maintaining answer specificity. SA was 1.0000 in both Flow C and Flow D, confirming that the major structural elements required for evaluation were generated without omission in the ontology-structuring stage.

E2E-RT was shortest for Flow A at 11.19 seconds, while Flow D recorded 15.06 seconds. Although Flow D includes ontology mapping, REG-based retrieval control, and RAG retrieval, it showed a shorter processing time than Flow C. This suggests that REG-based retrieval control can contribute to processing-path optimization by constraining data-access paths according to query type and reducing unnecessary retrieval.

## 6. Discussion

### 6.1 Interpretation of Experimental Results

Overall, the final proposed architecture, Flow D, can be viewed as a structure that strengthens the use of evidence aligned with query intent by sequentially performing ontology-based query structuring, REG-based retrieval control, and RAG-based evidence retrieval in academic administration question answering. In particular, Flow D achieved the highest final decision accuracy and generated the major structural elements required for ontology structuring without omission. Therefore, in domains such as academic administration where data sources such as regulations, schedules, and notices are separated, a RAG architecture that combines ontology-based query structuring with REG-based retrieval control may be more suitable than an LLM-only architecture.

## 6.2 Remaining Error Factors

However, the proposed architecture did not completely eliminate errors. First, if the domain, entity, or condition is incorrectly mapped during the process of converting a user query into an ontology structure, the error can propagate to subsequent retrieval and response-generation stages. Second, if documents not directly related to the query are retrieved in the RAG stage or if necessary evidence is omitted, the LLM may generate a response based on incomplete evidence. Third, although REG rules are effective for selecting data sources by query type and rewriting retrieval queries, they have limitations in fully covering exceptional clauses or regulatory interpretations that are difficult to quantify. Fourth, in the final response-generation stage, overgeneralization in expression may occur when the LLM reconstructs retrieval results or structuring results in natural language.

## 6.3 Applicability of the Proposed Framework

The proposed architecture extends an academic administration question-answering system beyond a simple natural-language response-generation system into an integrated structure that performs query interpretation, data-source selection, evidence retrieval, and response generation. Although Flow D proposed in this study was designed based on K University's academic administration data, it can be extended to academic administration services at other universities or to regulation-based administrative question-answering systems by reconstructing the domain ontology, REG-based retrieval-control rules, and data sources.

## 7. Conclusion

In this study, four workflows with different levels of component integration, Flow A through Flow D, were designed to automate university academic administration question answering, and the effects of the final proposed architecture, Flow D, on the decision accuracy and response stability of academic administration question answering were analyzed.

The experimental results showed that the final proposed architecture, Flow D, achieved the highest final decision accuracy among the comparison targets with a DCR of 0.6923 and maintained an SA of 1.0000. This suggests that the combination of ontology-based query structuring, REG-based retrieval control and response guidance, and RAG-based evidence retrieval can support the use of evidence aligned with query intent and contribute to improved decision accuracy in academic administration question answering.

Nevertheless, this study has limitations in that it used an experimental set centered on representative queries, and the ontology and REG rules depend on manual design. In addition, REG in this study was implemented not as a complete rule-inference engine that independently determines all administrative conditions, but as a control layer that guides data-source selection by query type, retrieval-query rewriting, and evidence-based response generation. Future research should expand the query set based on actual user logs, automatically generate and update the ontology and REG rules, evaluate the reliability of retrieval results, and verify evidence consistency after response generation.

## Acknowledgment

The research was supported by the Regional Innovation System & Education(RISE) program through the Gyeonggi RISE Center, funded by the Ministry of Education(Moe) and Gyeonggi-do, Republic of

Korea.(2026-RISE-09-002).

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